Crime and What Else?

Investigating the Predictive Power of Social Factors on Incarceration Rates

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1 Title, Team, and Affiliations

Project title: *Crime and What Else? Investigating the Predictive Power of Social Factors on Incarceration Rates*

Name	Role	Willamette email
Emery Kerani	Owner Product Lead	eskerani@willamette.edu

Peer Stakeholder POs: Mary Rose Krouse, Shanti Brodnick, Rohan Srinivas Babu

Repository: `https://github.com/eskerani/prison-predictor-capstone/tree/main`

2 Abstract

(150 to 250 words after you have results. Summarize: problem, approach, key finding, impact, limitation. Write this section last.)

3 Introduction: Problem and Research Questions

Problem statement: The United States prison system currently holds nearly 2 million people, and for decades has imprisoned its inhabitants at one of the highest rates in the world (Lurigio, 2024). A variety of legal factors, including mandatory minimum sentences and "three strikes" laws, have contributed to the unprecedented rise in incarceration in the US since the 1970s and 80s (DeFina and Hannon, 2013; Lurigio, 2024). The US incarceration rate has since levelled off, remaining relatively steady since the 2010s; however, this does not reflect the nation's crime rate, which has fallen since peaking in in the 1990s (Pettit and Gutierrez, 2013). Given that the US's criminal legal system wields tremendous power—power that is most often turned against already disadvantaged individuals and their communities—it is important that we recognize protective and risk factors for incarceration, many of which are social in nature (Lurigio, 2024; Henry, 2020; Pettit and Gutierrez, 2013). This project seeks to interrogate the dominant narrative that crime is the primary cause of incarceration by examining three other risk factors.

Primary research question: How do the predictive power of social factors (such as poverty, race, or government spending on police) compare to that of violent and property crime rates when using support vector regression to model the national incarceration rate?

Secondary questions (if any): If crime is in fact the best predictor of incarceration rates, which social factor is the best predictor of crime rates?

Changes since M1 proposal: None yet

4 Impact and Stakeholders

The key stakeholders in this work are those who are most often targeted by criminalization and incarceration, primarily poor and BIPOC folks (Pettit and Gutierrez, 2013). Additionally, communities across the country are continually harmed by increasing incarceration rates, as incarceration has been shown to negatively impact both community health (Wildeman and Wang, 2017) and poverty (DeFina and Hannon, 2013). By critically examining the circumstances that can contribute to incarceration, this project seeks to uplift the work of organizations such as Transformative Justice Community and the Oregon Justice Resource Center, which advocate for solutions to harm that shift our paradigm away from imprisonment.

My shared theme is social justice and identity-driven disparity analysis. As others who are interested in social justice but perhaps not as familiar with the prison-industrial complex as I am, I hope that they'll point out anywhere I make assumptions/lack context and help ensure that my project doesn't steer anyone toward misinformed conclusions.

5 Related Work and Background

Wildfire smoke and respiratory health are well studied at regional scale (Reid et al. 2016; Liu et al. 2016). This project applies county-level public data to Oregon specifically.

6 Data and Data Engineering

Sources:

Source	Role	Access

Engineering summary: Raw data in `data/raw/`; processed panel in `data/processed/`; ingest scripts in `src/ingest/`.

```r
# Replace with your project data when ready:
# panel <- readRDS(here::here("data/processed/analysis_panel.rds"))
# nrow(panel)
knitr::kable(data.frame(
  check = c("Rows", "Counties", "Date range"),
  status = c("stub", "stub", "stub")
))
```

Table 3: Example QC summary (replace with your panel)

check	status
Rows	stub
Counties	stub
Date range	stub

7 Ethics and Responsible Use

- **Privacy:**
- **Fairness / equity:**
- **Limitations of setting:**
- **Non-causal framing:** *(if observational)*

8 Methods and Evaluation

Design:

Primary analysis:

Evaluation metrics:

Train / validation / holdout:

9 Results and Visualizations

(Stub: key findings with Figure 1 and tables.)

10 Discussion, Limitations, and Future Work

Interpretation:

Limitations: *(tie to actual data and design)*

Future work:

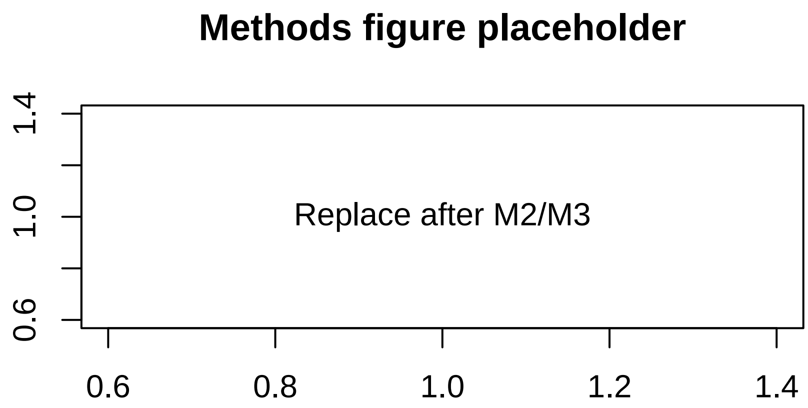

Figure 1: Replace with a methods diagram or exploratory plot

11 Conclusions and Recommendations

(Short, stakeholder-facing takeaways.)

12 Reproducibility and References

Reproduce main results:

```
cd your-repo
quarto render deliverables/M4-writeup-draft/writeup.qmd
```

Dependencies: `requirements.txt` or `renv.lock` in repo root.

Data: URLs and approval status documented in `docs/data-dictionary.md`.

13 References

Liu, J. C., L. J. Mickley, M. P. Sulprizio, et al. 2016. "Wildfire-Specific Fine Particulate Matter and Risk of Hospital Admissions in Urban and Rural Counties." *Epidemiology* 27 (3): 350–57.

Reid, C. E., M. Brauer, F. H. Johnston, et al. 2016. "Critical Review of Health Impacts of Wildfire Smoke Exposure." *Environmental Health Perspectives* 124 (8): 1334–43.